Separating methane growth from changes in its sources and sinks

Methane surge reconstruction

Following equation (1) of He et al. (2026, doi:10.1126/sciadv.adz9007), the annual global methane budget is

$$G_j = E_j - (k_{\mathrm{OH},j} + k_{\mathrm{other},j})m_j - L_j, \tag{1}$$

in which G_j is atmospheric mass growth, E_j is gross emission, L_j is positive soil uptake (all in Tg CH_4 yr^{-1}), m_j is atmospheric methane mass (Tg CH_4), and the k terms are chemical loss frequencies (yr^{-1}). The subscript 0 denotes 2019 and every Δ denotes the value in year j minus its 2019 value. Soil uptake must not also be included in k_{other}.

Writing $k_0 = k_{\mathrm{OH},0} + k_{\mathrm{other},0}$ and expanding gives

$$G_j = \underbrace{G_0 - k_0\Delta m_j}_{R_j} + \underbrace{\Delta E_j}_{F_{E,j}} \underbrace{-m_0\Delta k_{\mathrm{OH},j}}_{F_{\mathrm{OH},j}} + \epsilon_j, \tag{2}$$

$$\epsilon_j = -m_0\Delta k_{\mathrm{other},j} - (\Delta k_{\mathrm{OH},j} + \Delta k_{\mathrm{other},j})\Delta m_j - \Delta L_j. \tag{3}$$

He et al. neglect ϵ_j. The implementation retains it as a diagnostic, separately from the three published attribution terms. R_j is the residual growth under baseline rates evaluated at the actual burden. It is a growth contribution, and is therefore generally nonzero in 2019. Its anomaly is $-k_0\Delta m_j$. The forced growth anomaly is

$$F_j = \Delta G_j + k_0\Delta m_j \simeq F_{E,j} + F_{\mathrm{OH},j}. \tag{4}$$

The positive adjustment $k_0\Delta m_j$ removes the extra loss caused by methane accumulation. A percentage change in an arbitrarily weighted OH mean is not sufficient to calculate Δk_{OH}. The chemical lifetime must be methane-reaction weighted.

For a published OH removal flux $Q_j = k_{\mathrm{OH},j}m_j$, the exact conversion to the baseline-burden OH forcing is

$$F_{\mathrm{OH},j} = \frac{-\Delta Q_j + k_{\mathrm{OH},0}\Delta m_j}{1 + \Delta m_j/m_0}. \tag{5}$$

Thus, a change in removal flux must not be substituted directly for the change in loss frequency. This conversion requires the burden and baseline removal rate, and will not be made where those are unavailable.

These equations apply to globally integrated budgets. Regional methane growth also includes transport, so no regional relaxation adjustment will be inferred from local emissions alone. Published regional emission anomalies can be compared separately. No inversion is performed.